MODEL EXIT EXAM 2026 — QUESTIONS & ANSWER SHEET

PART 1: QUESTIONS

Q1. In a CIDR block of 192.168.100.0/23, what is the broadcast address?

a. 192.168.102.0

b. 192.168.100.1

c. 192.168.100.255

d. 192.168.101.255

Q2. Which of the following describes the possible alternative ways in which files on a disk can be accessed?

a. serial or random

b. object or serial

c. text or binary

d. binary or random

e. random or binary

Q3. In Agile development, what is a sprint?

a. A testing phase

b. A debugging session

c. A time-boxed development cycle

d. A rapid design process

Q4. Decision trees classify data points by recursively splitting the data space based on _____ features.

a. numerical

b. most relevant

c. most correlated

d. all available

Q5. What is meant by software quality in software engineering?

a. Complexity of code

b. High number of features

c. Degree to which software meets requirements and is free of defects

d. Low cost

Q6. Which class of IP addresses is reserved for experimental use?

a. Class A

b. Class E

c. Class C

d. Class D

Q7. Which of the following search algorithms guarantees both completeness and optimality, assuming the heuristic is admissible and consistent?

a. Greedy Best-First Search

b. A* Search

c. Depth-First Search

d. Hill Climbing

Q8. Which protocol uses port 443 and operates at Layer 7?

a. FTP

b. HTTPS

c. SSH

d. SNMP

Q9. Consider the following method, setValue, which adds a value into an array list: public void setValue(int position, String valueIn) { list[position-1] = valueIn; } Assuming that list has been declared as an array of Strings, which one of the following might cause the setValue method to throw an ArrayIndexOutOfBounds exception:

a. the type of the parameter position being declared as an int

b. not declaring the method static

c. the return type being declared as void

d. the value of the first parameter

Q10. Which Activity lifecycle callback is the first one called after the Activity is launched and is where initial setup like inflating the UI layout typically occurs?

a. onStart()

b. onCreate()

c. onResume()

d. onAttach()

Q11. Which CSS property is used to control the space between the content and the border of an element?

a. padding

b. margin

c. spacing

d. border-spacing

Q12. Which of the following is known as a set of entities of the same type that share same properties, or attributes?

a. Entity set

b. Relation set

c. Tuples

d. Entity Relation model

Q13. In the TCP/IP model, which layer corresponds to the Transport layer of OSI?

a. Transport

b. Application

c. Network Access

d. Internet

Q14. The values appearing in given attributes of any tuple in the referencing relation must likewise occur in specified attributes of at least one tuple in the referenced relation. What integrity constraint dictates this?

a. Referencing

b. Referential

c. Specific

d. Primary

Q15. Which of the following is not a Boolean expression?

a. 10 == 2*5

b. 20 = 5+15

c. 10**2 != 100

d. True != True

Q16. Which of the following data structures uses the LIFO (Last In First Out) principle?

a. Binary Tree

b. Stack

c. Linked List

d. Queue

Q17. What will be the output of the following code? var x = 5; var y = '5'; console.log(x == y);

a. true

b. undefined

c. null

d. false

Q18. Consider the following method, setValue (same as Q9). Assuming that list has been declared as an array of Strings, which one of the following might cause the setValue method to throw an ArrayIndexOutOfBounds exception:

a. not declaring the method static

b. the type of the parameter position being declared as an int

c. the value of the first parameter

d. the method having more than one parameter

e. the return type being declared as void

Q19. What will be printed when the following code is executed? >>> tuple1 = (1, 2, 3, 4) >>> print(tuple1[1:-1])

a. (2, 3)

b. Error: tuple slicing

c. (2, 3, 4)

d. (1, 2, 3, 4)

Q20. In COCOMO model, Initial estimate Ei = ___________

a. a*(KLOC)*b

b. b*(KLOC)*a

c. a(KLOC)b

d. b(KLOC)a

Q21. A process goes from running-state to blocked-state whenever a running process:

a. moved to secondary storage because of memory shortage

b. None

c. interrupted by a higher priority process

d. CPU executes context switching

e. needs to get input/output in the middle of execution

Q22. Which of the following is not true about functions in python?

a. Functions are reusable pieces of program

b. Functions provide better modularity for your programs

c. Function definitions can alter flow of execution

d. Python allows user defined functions

Q23. Which Android UI component is described as a 'microactivity' that represents a behavior or portion of the UI and must be hosted within an Activity?

a. View

b. Adapter

c. Fragment

d. Service

Q24. Which method is used to add an element to the end of an array in JavaScript?

a. pop()

b. push()

c. shift()

d. unshift()

Q25. Four conditions are required for a deadlock to occur. Which one of the following is NOT a condition of deadlock?

a. Circular wait

b. Hold and wait

c. Mutual exclusion

d. No preemption

e. Busy wait

Q26. Mutual execution algorithms are primarily used to resolve:

a. None

b. Busy-waiting

c. Deadlock

d. Starvation

e. Race condition

Q27. HMMs are useful for modeling sequential data where the underlying states are ___.

a. Pre-defined

b. partially observable

c. always observable

d. completely random

Q28. What will be printed when the following code is executed? >>> tuple1 = (1, 2, 3, 4) >>> print(tuple1[1:-1]) (same as Q19)

a. (2, 3)

b. Error: tuple slicing

c. (2, 3, 4)

d. (1, 2, 3, 4)

Q29. In COCOMO model, Initial estimate Ei = ___________ (same as Q20)

a. a*(KLOC)*b

b. b*(KLOC)*a

c. a(KLOC)b

d. b(KLOC)a

Q30. Which of the following is true regarding the flow of execution in a while statement?

a. If the condition is yielding not true, execute the body

b. If the condition is false, exit the while statement and continue execution

c. All

d. Evaluates the condition yielding only True

Q31. If an algorithm has a time complexity of $O(n^2)$, what happens to the running time if the input size is doubled?

a. It triples

b. It doubles

c. It quadruples

d. It remains the same

Q32. Which one of the following scheduling algorithms most likely to cause starvation?

a. Round-Robin (RR)

b. First-Come, First Served (FCFS)

c. Short-Job First (SJF)

d. None

e. Priority-Based With Aging

Q33. Bereket needs to send an email to his supervisor with an attachment that includes sensitive information. He wants to maintain the confidentiality of this information. Which of the following choices is the BEST choice to meet his needs?

a. Digital signature

b. Encryption

c. Hashing

d. Data masking

Q34. The main purpose of STP (Spanning Tree Protocol) is to:

a. Prioritize traffic

b. Monitor SNMP traps

c. Prevent Layer 2 loops

d. Filter traffic by VLAN

Q35. Consider the following setValue method (same as Q9/Q18). Which one might cause an ArrayIndexOutOfBounds exception:

a. not declaring the method static

b. the type of the parameter position being declared as an int

c. the value of the first parameter

d. the method having more than one parameter

e. the return type being declared as void

Q36. Which command can display the current ARP cache on a Windows system?

a. ipconfig

b. ping

c. arp -a

d. netstat -r

Q37. In a graph, what does a cycle represent?

a. A linear path

b. A spanning tree

c. A set of disconnected vertices

d. A path that starts and ends at the same vertex

Q38. Which of the following is a divide and conquer algorithm?

a. Binary Search

b. Insertion Sort

c. Linear Search

d. Bubble Sort

Q39. Which of the following is true regarding the flow of execution in a while statement? (same as Q30)

a. If the condition is yielding not true, execute the body

b. If the condition is false, exit the while statement and continue execution

c. All

d. Evaluates the condition yielding only True

Q40. Which one of the following scheduling algorithms most likely to cause starvation? (same as Q32)

a. Round-Robin (RR)

b. First-Come, First Served (FCFS)

c. Short-Job First (SJF)

d. None

e. Priority-Based With Aging

Q41. Which one of the following scheduling algorithms most likely to cause starvation? (same as Q32)

a. Round-Robin (RR)

b. First-Come, First Served (FCFS)

c. Short-Job First (SJF)

d. None

e. Priority-Based With Aging

Q42. Bereket needs to send an email to his supervisor with an attachment that includes sensitive information. (same as Q33)

a. Digital signature

b. Encryption

c. Hashing

d. Data masking

Q43. Consider the following setValue method — which one might cause an ArrayIndexOutOfBounds exception:

a. not declaring the method static

b. the type of the parameter position being declared as an int

c. the value of the first parameter

d. the method having more than one parameter

e. the return type being declared as void

Q44. Which of the following best describes a compiled language?

a. Code is always faster in compiled languages because they have no errors

b. Code is executed line-by-line

c. Code is translated directly into machine code before execution

d. Compiled languages cannot be used for low-level programming

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Q45. Which principle refers to minimizing the interdependence between software modules?

a. Modularity

b. Cohesion

c. Abstraction

d. Coupling

Q46. _____ is a set of one or more attributes taken collectively to uniquely identify a record.

a. Primary Key

b. Foreign key

c. Super key

d. Candidate key

Q47. Consider the inheritance hierarchy shown in the diagram below: [Class inheriting diagram with two subclasses]. What does the arrow from subclasses to Class represent?

a. Inheritance

b. Composition

c. Aggregation

d. Association

Q48. You are comparing different types of authentications. Of the following choices, which one uses multifactor authentication?

a. A system that requires users to enter a username and password

b. A system that requires users to have a smart card and a PIN

c. A system that checks an employee's fingerprint and does a vein scan

d. A cipher door lock that requires employees to enter a code to open the door

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Q49. Which evaluation metric is most suitable for imbalanced classification problems?

a. Accuracy

b. R-squared

c. F1-Score

d. Mean Squared Error (MSE)

Q50. In linear regression, the predicted value is a linear function of the _____.

a. distance from the hyperplane

b. independent variables

c. class probabilities

d. dependent variable

Q51. Your organization hired a cybersecurity expert to perform a security assessment. After running a vulnerability scan, she sees: Host IP 192.168.1.10 OS Apache httpd 2.433 Vulnerable to mod_auth exploit. However, she verified that the mod_auth module has not been installed or enabled on the server. Which of the following BEST explains this scenario?

a. A false positive

b. The result of a credentialed scan

c. The result of a non-credentialed scan

d. A false negative

Q52. What is the main function of the three-way handshake in TCP?

a. Flow control

b. Authentication

c. Reliable data delivery

d. Connection setup and synchronization

Q53. A class called Network has been declared with private attributes maxNumberOfUsers and currentNumberOfUsers. A class SubNet extends Network. In a program that declares an initializes an object sub of SubNet, which of the following statements would NOT cause a compiler error?

a. sub.maxNumberOfUsers = 100;

b. sub.setMaxNumberOfUsers(100);

c. sub.currentNumberOfUsers = 50;

d. System.out.println(sub.maxNumberOfUsers);

Q54. Which one of the following mechanisms allows execution of multiple processes by switching CPU even if the currently executing process is not completed?

a. Time-sharing

b. Multiprogramming

c. Multiprocessing

d. None

e. Monoprogramming

Q55. _____ involves generating, collecting, disseminating, and storing information.

a. Concurrent Management

b. Critical Management

c. Configuration Management

d. Communication Management

Q56. Chala is a security practitioner tasked with ensuring that the information on the organization's public website is not changed by anyone outside the organization. This task is an example of ensuring _____.

a. Availability

b. Integrity

c. Confidentiality

d. Authentication

Q57. Which software process model emphasizes strict sequential flow between phases?

a. Agile Model

b. Incremental Model

c. Spiral Model

d. Waterfall Model

Q58. What is the output of the following code? >>> my_list = [1,2,3,2,4] >>> my_list.remove(2) >>> my_list

a. None

b. [1,2,2,4]

c. [1,3,2,4]

d. [1,3,4]

Q59. Which of the following best describes the role of a file system in an operating system?

a. It manages the computer's hardware resources, such as CPU and memory

b. It provides a structured way to store, organize, retrieve, and manage files on storage devices

c. It optimizes network bandwidth usage for file transfers over the internet

d. It handles user authentication and access control for logging into the system

e. None

Q60. Delphi approach is also called _____ group consensus technique.

a. Fully Consultative

b. Non-Consultative

c. Consultative

d. Partially Consultative

Q61. What is the output of the following code? >>> my_list = [1,2,3,4,5,6,7,8,9,10] >>> my_list[5::-2]

a. [6,4,2]

b. [5,7,9]

c. [5,4,3,2,1]

d. []

Q62. _____ is a hierarchical and incremental decomposition of the project into phases, deliverables and work packages.

a. Work Integration Structure

b. Task Integration Structure

c. Bottom-up Integration Structure

d. Work Breakdown Structure

Q63. The bias-variance trade-off implies that:

a. Increasing model complexity reduces both bias and variance

b. Underfitting is caused by low variance

c. Simple models have high bias and low variance

d. Overfitting occurs when bias is too high

Q64. What is the primary objective of virtual memory in a computer system?

a. increase the physical memory capacity

b. None

c. increase execution speed by catching frequently used data

d. allow processes use more memory than physical available

e. secure part of memory from being accessed by other processes

Q65. What is printed by the following statements? >>> str = 'This is python code' >>> print(str[2] + str[-6])

a. tn

b. io

c. in

d. to

Q66. Sara is a database manager. Sara is allowed to add new users to the database, remove current users, and create new usage functions for the users. Sara is not allowed to read the data in the fields of the database itself. This is an example of:

a. Role Based Access Control

b. Discretionary Access Control

c. Mandatory Access Control

d. Behavior Based Access Control

Q67. Which type of data can be stored in the database?

a. Text, files containing data

b. All of the above

c. Image oriented data

d. Data in the form of audio or video

Q68. What is the primary function of an Intent in Android app navigation?

a. To manage background threads for long-running operations

b. To request an action from another app component

c. To store application settings persistently

d. To define the visual appearance of a screen

Q69. What is the primary purpose of WAI-ARIA?

a. To improve website SEO rankings

b. To make web applications more accessible

c. To increase website loading speed

d. To replace HTML with a new accessibility standard

Q70. A program is required to add two integer numbers together. The GUI for this program shows two text fields and a Calculate button. What is the component labeled with an arrow in the diagram?

a. JFrame

b. JTextField

c. JLabel

d. JButton

Q71. What is meant by software quality in software engineering? (same as Q5)

a. Complexity of code

b. High number of features

c. Degree to which software meets requirements and is free of defects

d. Low cost

Q72. The specific set of sequential tasks upon which the project completion date depends or the longest full path is called _____.

a. Full path

b. Project Path

c. Critical Path

d. Complete Path

Q73. Which of the following sorting algorithms has the best average-case time complexity?

a. Quick Sort

b. Bubble Sort

c. Insertion Sort

d. Selection Sort

Q74. What is the worst-case time complexity of Merge Sort?

a. $O(n^2)$

b. O(log n)

c. O(n log n)

d. O(n)

Q75. How do you check if an element exists in a set? Using

a. in keyword

b. exists()

c. contains()

d. check()

Q76. Which of the following is an example of a 'Something you know' authentication factor?

a. GPS

b. Finger Printing

c. Password

d. ATM Card

Q77. What is the output of this expression: >>> (2 + 5) ** 2 // 10 + 2 * 3

a. 12.0

b. 10

c. 8

d. 12

Q78. A technique to prevent modification anomalies by creating a different class of relations in RDMS are called?

a. Functional Dependencies

b. Normal Forms

c. Referential integrity

d. Data integrity

Q79. Which operation is used to extract specified columns from a table?

a. Substitute

b. Extract

c. Project

d. Join

Q80. A team is designing a weather app that must work offline in remote areas but also sync data when internet connectivity is available. Which architectural approach addresses this need?

a. Only online-based architecture

b. Peer-to-peer architecture

c. Client-side caching with periodic synchronization

d. Pure serverless architecture (no offline support)

Q81. Which of the following is the primary goal of software engineering?

a. To produce reliable, efficient, and maintainable software

b. To develop low-level machine code

c. To increase hardware performance

d. To create visually appealing user interfaces

Q82. Which of the following is a divide and conquer algorithm? (same as Q38)

a. Binary Search

b. Insertion Sort

c. Linear Search

d. Bubble Sort

Q83. In the context of search problems, what does a heuristic function do?

a. Estimates the cost from a node to the goal

b. Calculates the actual cost of a path

c. Calculates the cost from start to goal

d. Finds the goal state directly

Q84. How do you create a CSS rule that applies only to screen devices with a maximum width of 600px?

a. @media screen and (min-width: 600px) { ... }

b. @media print screen and (max-width: 600px) { ... }

c. @media screen and (max-width: 600px) { ... }

d. @media all and (max-width: 600px) { ... }

Q85. What is the main advantage of using kernel methods in machine learning?

a. They are faster than other machine learning algorithms for all tasks

b. They can only be used with Support Vector Machines (SVMs)

c. They require less data for training compared to other methods

d. They allow linear algorithms to work effectively with non-linear data

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Q86. Which CSS property is used to control the stacking order of positioned elements?

a. display

b. position

c. float

d. z-index

Q87. How can web developers accommodate users with mobility impairments?

a. Ensuring all controls are accessible via keyboard

b. Making users rely on a mouse for navigation

c. Limiting access to only touch-screen devices

d. Removing all interactive elements

Q88. Consider class ClassA (with testMethod printing 'Hello') and class ClassB extends ClassA (with testMethod printing 'Goodbye'). If a ClassB object calls testMethod(), what is printed?

a. Hello

b. Goodbye

c. Hello then Goodbye

d. Goodbye then Hello

Q89. Your organization hired a cybersecurity expert. After a vulnerability scan: Host IP 192.168.1.10 OS Apache httpd 2.433 Vulnerable to mod_auth exploit. She verified that mod_auth has NOT been installed.

Which BEST explains this scenario?

a. The result of a credentialed scan

b. A false negative

c. The result of a non-credentialed scan

d. A false positive

Q90. What is printed by the following statements? >>> Str1 = 'PYTHON programming' >>> print(Str1[2] * Str1.index('O'))

a. OOOO

b. THON

c. TTTT

d. YYYY

Q91. What is the output of the following code? >>> my_list = [1,2,3,4,5,6,7,8,9,10] >>> my_list[5::-2] (same as Q61)

a. [6,4,2]

b. [5,7,9]

c. [5,4,3,2,1]

d. []

Q92. Which statement about ensemble methods (e.g., Random Forests) is TRUE?

a. They require all base models to be identical

b. They perform poorly compared to single models

c. They combine weak learners to reduce bias only

d. They average predictions to decrease variance

Q93. What is the primary purpose of the RecyclerView widget?

a. To display a single image on the screen

b. To display lists of data by efficiently reusing item views

c. To provide a simple button with click functionality

d. To manage background tasks efficiently

Q94. When you want to navigate to a specific Activity within your application, which type of Intent is typically used?

a. Explicit intent

b. Implicit intent

c. Pending intent

d. Broadcast intent

Q95. What is the purpose of the box-sizing property in CSS?

a. To include padding and border in the element's total size

b. To set the total margin of the box

c. To define the size of the total content and margin box

d. To apply a shadow and margin to the box

Q96. Your organization's backup policy for a file server dictates that the amount of time needed to restore backups should be minimized. Which of the following backup plans would BEST meet this need?

a. Full backups on Sunday and incremental backups on the other six days of the week

b. Full backups on Sunday and differential backups on the other six days of the week

c. Differential backups on Sunday and incremental backups on the other six days of the week

d. Incremental backups on Sunday and differential backups on the other six days of the week

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Q97. In a neural network, the activation function introduces:

a. Regularization to prevent overfitting

b. Non-linearity to model complex patterns

c. Dimensionality reduction

d. Linear decision boundaries

Q98. A _____ is the lowest level of work on the project.

a. Task Set

b. Task

c. Milestone

d. Work product

Q99. Return of Investment (ROI) ROI = _____

a. (Gain - Cost)/Gain

b. (Cost - Gain)/Gain

c. (Gain - Cost)/Cost

d. (Cost - Gain)/ Cost

Q100. A startup is building an e-commerce platform expecting rapid growth. They need an architecture that allows independent scaling of the product catalog, payment processing, and recommendation engine. Which architectural style is most suitable?

a. Single-tier architecture (all logic on the client side)

b. Monolithic architecture (all components in one codebase)

c. Shared-database architecture (all services use one database)

d. Microservices architecture (independent, loosely coupled services)

Q101. Slack = _____

a. LS-EF

b. ES-LF

c. ES-LS

d. LF-EF

Q102. Which of the following is the correct syntax to create a function in JavaScript?

a. function: myFunction() {}

b. function = myFunction() {}

c. function => myFunction() {}

d. function myFunction() {}

Q103. Which of the following function declarations is correct to define a function that accepts a variable number of arguments?

a. def func(*args):

b. Not possible in python

c. def func(args*):

d. def func(**args):

Q104. Which of the following is a key challenge in reinforcement learning?

a. Requirement of perfectly balanced datasets

b. The trade-off between exploration and exploitation

c. Lack of labeled training data

d. High computational cost of matrix operations

Q105. Which algorithm is most appropriate when the cost of actions varies and finding the least-cost solution is important?

a. Depth-First Search

b. Breadth-First Search

c. Greedy Best-First Search

d. Uniform Cost Search

Q106. How can you apply a CSS Grid layout to a container?

a. display: block;

b. display: grid;

c. display: inline-grid;

d. display: flex;

PART 2: ANSWER SHEET WITH EXPLANATIONS

Q1. In a CIDR block of 192.168.100.0/23, what is the broadcast address?

a. 192.168.102.0

b. 192.168.100.1

c. 192.168.100.255

✔ d. 192.168.101.255

Correct Answer: d. 192.168.101.255

Explanation: A /23 block covers 192.168.100.0–192.168.101.255. The broadcast address is the last IP: 192.168.101.255.

Q2. Which of the following describes the possible alternative ways in which files on a disk can be accessed?

✔ a. serial or random

b. object or serial

c. text or binary

d. binary or random

e. random or binary

Correct Answer: a. serial or random

Explanation: Files on a disk can be accessed either sequentially (serial) from start, or directly (random) by jumping to any location.

Q3. In Agile development, what is a sprint?

a. A testing phase

b. A debugging session

✔ c. A time-boxed development cycle

d. A rapid design process

Correct Answer: c. A time-boxed development cycle

Explanation: In Agile/Scrum, a sprint is a fixed-length iteration (usually 1–4 weeks) during which a working increment of the product is developed.

Q4. Decision trees classify data points by recursively splitting the data space based on _____ features.

a. numerical

b. most relevant

✔ c. most correlated

d. all available

Correct Answer: c. most correlated

Explanation: Decision trees split using the most correlated (most informative) features that best separate classes.

Q5. What is meant by software quality in software engineering?

a. Complexity of code

b. High number of features

✔ c. Degree to which software meets requirements and is free of defects

d. Low cost

Correct Answer: c. Degree to which software meets requirements and is free of defects

Explanation: Software quality measures how well software satisfies functional and non-functional requirements and is reliable/defect-free.

Q6. Which class of IP addresses is reserved for experimental use?

a. Class A

✔ b. Class E

c. Class C

d. Class D

Correct Answer: b. Class E

Explanation: Class E (240.0.0.0–255.255.255.255) is reserved for experimental and research purposes by IANA.

Q7. Which of the following search algorithms guarantees both completeness and optimality, assuming the heuristic is admissible and consistent?

a. Greedy Best-First Search

✔ b. A* Search

c. Depth-First Search

d. Hill Climbing

Correct Answer: b. A* Search

Explanation: A* is both complete and optimal when the heuristic is admissible (never overestimates) and consistent.

Q8. Which protocol uses port 443 and operates at Layer 7?

a. FTP

✔ b. HTTPS

c. SSH

d. SNMP

Correct Answer: b. HTTPS

Explanation: HTTPS operates at Layer 7 (Application layer) and uses port 443, providing encrypted HTTP communication via TLS/SSL.

Q9. Consider the following method, setValue, which adds a value into an array list: public void setValue(int position, String valueIn) { list[position-1] = valueIn; } Assuming that list has been declared as an array of Strings, which one of the following might cause the setValue method to throw an ArrayIndexOutOfBounds exception:

a. the type of the parameter position being declared as an int

b. not declaring the method static

c. the return type being declared as void

✔ d. the value of the first parameter

Correct Answer: d. the value of the first parameter

Explanation: If position = 0, then list[position-1] = list[-1] accesses the last element. If position is negative or out of bounds, it throws an exception. The value is the risk.

Q10. Which Activity lifecycle callback is the first one called after the Activity is launched and is where initial setup like inflating the UI layout typically occurs?

a. onStart()

✔ b. onCreate()

c. onResume()

d. onAttach()

Correct Answer: b. onCreate()

Explanation: onCreate() is the first lifecycle callback called after an Activity is launched. It is where you initialize the UI by inflating layouts.

Q11. Which CSS property is used to control the space between the content and the border of an element?

✔ a. padding

b. margin

c. spacing

d. border-spacing

Correct Answer: a. padding

Explanation: The CSS 'padding' property controls the space between an element's content and its border (inside the border).

Q12. Which of the following is known as a set of entities of the same type that share same properties, or attributes?

✔ a. Entity set

b. Relation set

c. Tuples

d. Entity Relation model

Correct Answer: a. Entity set

Explanation: An entity set is a collection of entities of the same type sharing the same attributes (e.g., all employees).

Q13. In the TCP/IP model, which layer corresponds to the Transport layer of OSI?

✔ a. Transport

b. Application

c. Network Access

d. Internet

Correct Answer: a. Transport

Explanation: In the TCP/IP model, the Transport layer directly corresponds to the OSI Transport layer — both handle end-to-end communication (TCP/UDP).

Q14. The values appearing in given attributes of any tuple in the referencing relation must likewise occur in specified attributes of at least one tuple in the referenced relation. What integrity constraint dictates this?

a. Referencing

✔ b. Referential

c. Specific

d. Primary

Correct Answer: b. Referential

Explanation: Referential integrity ensures that every foreign key value in one table corresponds to an existing primary key in the referenced table.

Q15. Which of the following is not a Boolean expression?

a. 10 == 2*5

✔ b. 20 = 5+15

c. 10**2 != 100

d. True != True

Correct Answer: b. 20 = 5+15

Explanation: In Python, '=' is assignment, not comparison. A single '=' is not a Boolean expression. '==' is required.

Q16. Which of the following data structures uses the LIFO (Last In First Out) principle?

a. Binary Tree

✔ b. Stack

c. Linked List

d. Queue

Correct Answer: b. Stack

Explanation: A Stack uses LIFO — the last element pushed is the first one popped. A Queue uses FIFO.

Q17. What will be the output of the following code? var x = 5; var y = '5'; console.log(x == y);

✔ a. true

b. undefined

c. null

d. false

Correct Answer: a. true

Explanation: JavaScript's == operator performs type coercion. 5 == '5' is true because '5' is coerced to the number 5 before comparison.

Q18. Consider the following method, setValue (same as Q9). Assuming that list has been declared as an array of Strings, which one of the following might cause the setValue method to throw an ArrayIndexOutOfBounds exception:

a. not declaring the method static

b. the type of the parameter position being declared as an int

✔ c. the value of the first parameter

d. the method having more than one parameter

e. the return type being declared as void

Correct Answer: c. the value of the first parameter

Explanation: If position is 0 or negative beyond the array bounds, then list[position-1] will cause an ArrayIndexOutOfBounds exception.

Q19. What will be printed when the following code is executed? >>> tuple1 = (1, 2, 3, 4) >>> print(tuple1[1:-1])

✔ a. (2, 3)

b. Error: tuple slicing

c. (2, 3, 4)

d. (1, 2, 3, 4)

Correct Answer: a. (2, 3)

Explanation: tuple1[1:-1] slices from index 1 to the second-last element (exclusive). tuple1 = (1,2,3,4), so indexes 1 to 2 gives (2,3).

Q20. In COCOMO model, Initial estimate Ei = ___________

a. a*(KLOC)*b

b. b*(KLOC)*a

✔ c. a(KLOC)b

d. b(KLOC)a

Correct Answer: c. a(KLOC)b

Explanation: In the COCOMO basic model, the effort equation is Ei = a*(KLOC)^b, often written as a(KLOC)b in compact notation.

Q21. A process goes from running-state to blocked-state whenever a running process:

a. moved to secondary storage because of memory shortage

b. None

c. interrupted by a higher priority process

d. CPU executes context switching

✔ e. needs to get input/output in the middle of execution

Correct Answer: e. needs to get input/output in the middle of execution

Explanation: A process moves from running to blocked when it makes an I/O request and must wait.

Q22. Which of the following is not true about functions in python?

a. Functions are reusable pieces of program

b. Functions provide better modularity for your programs

✔ c. Function definitions can alter flow of execution

d. Python allows user defined functions

Correct Answer: c. Function definitions can alter flow of execution

Explanation: Defining a function does not alter flow of execution — it merely defines a block of code. Calling a function alters flow.

Q23. Which Android UI component is described as a 'microactivity' that represents a behavior or portion of the UI and must be hosted within an Activity?

a. View

b. Adapter

✔ c. Fragment

d. Service

Correct Answer: c. Fragment

Explanation: A Fragment is described as a 'microactivity' in Android — it represents a reusable portion of UI or behavior hosted inside an Activity.

Q24. Which method is used to add an element to the end of an array in JavaScript?

a. pop()

✔ b. push()

c. shift()

d. unshift()

Correct Answer: b. push()

Explanation: In JavaScript, array.push(element) adds an element to the END of an array and returns the new length.

Q25. Four conditions are required for a deadlock to occur. Which one of the following is NOT a condition of deadlock?

a. Circular wait

b. Hold and wait

c. Mutual exclusion

d. No preemption

✔ e. Busy wait

Correct Answer: e. Busy wait

Explanation: The four standard deadlock conditions are: mutual exclusion, hold and wait, no preemption, and circular wait. Busy wait is NOT one of them.

Q26. Mutual execution algorithms are primarily used to resolve:

a. None

b. Busy-waiting

c. Deadlock

d. Starvation

✔ e. Race condition

Correct Answer: e. Race condition

Explanation: Mutual exclusion algorithms are used to avoid race conditions where multiple processes access shared resources unsafely.

Q27. HMMs are useful for modeling sequential data where the underlying states are ___.

a. Pre-defined

✔ b. partially observable

c. always observable

d. completely random

Correct Answer: b. partially observable

Explanation: In HMMs, the hidden states are not directly observable — only the emitted output symbols are.

Q28. What will be printed when the following code is executed? >>> tuple1 = (1, 2, 3, 4) >>> print(tuple1[1:-1]) (same as Q19)

✔ a. (2, 3)

b. Error: tuple slicing

c. (2, 3, 4)

d. (1, 2, 3, 4)

Correct Answer: a. (2, 3)

Explanation: tuple1[1:-1] slices from index 1 to the second-to-last (exclusive), giving (2, 3) from (1,2,3,4).

Q29. In COCOMO model, Initial estimate Ei = ___________ (same as Q20)

a. a*(KLOC)*b

b. b*(KLOC)*a

✔ c. a(KLOC)b

d. b(KLOC)a

Correct Answer: c. a(KLOC)b

Explanation: COCOMO initial effort estimate: Ei = a * (KLOC)^b, written as a(KLOC)b in function notation.

Q30. Which of the following is true regarding the flow of execution in a while statement?

a. If the condition is yielding not true, execute the body

✔ b. If the condition is false, exit the while statement and continue execution

c. All

d. Evaluates the condition yielding only True

Correct Answer: b. If the condition is false, exit the while statement and continue execution

Explanation: A while loop evaluates the condition before each iteration. If false, execution exits the loop.

Q31. If an algorithm has a time complexity of $O(n^2)$, what happens to the running time if the input size is doubled?

a. It triples

b. It doubles

✔ c. It quadruples

d. It remains the same

Correct Answer: c. It quadruples

Explanation: $O(n^2)$ means time grows as n squared. If n doubles to 2n, the time becomes $(2n)^2 = 4n^2$, i.e., it quadruples.

Q32. Which one of the following scheduling algorithms most likely to cause starvation?

a. Round-Robin (RR)

b. First-Come, First Served (FCFS)

✔ c. Short-Job First (SJF)

d. None

e. Priority-Based With Aging

Correct Answer: c. Short-Job First (SJF)

Explanation: SJF always picks the shortest available job, which can cause long jobs to wait indefinitely (starvation).

Q33. Bereket needs to send an email to his supervisor with an attachment that includes sensitive information. He wants to maintain the confidentiality of this information. Which of the following choices is the BEST choice to meet his needs?

a. Digital signature

✔ b. Encryption

c. Hashing

d. Data masking

Correct Answer: b. Encryption

Explanation: Encryption is the correct approach to maintain confidentiality of an email attachment.

Q34. The main purpose of STP (Spanning Tree Protocol) is to:

a. Prioritize traffic

b. Monitor SNMP traps

✔ c. Prevent Layer 2 loops

d. Filter traffic by VLAN

Correct Answer: c. Prevent Layer 2 loops

Explanation: STP (Spanning Tree Protocol) prevents switching loops in Layer 2 networks by blocking redundant paths.

Q35. Consider the following setValue method (same as Q9/Q18). Which one might cause an ArrayIndexOutOfBounds exception:

a. not declaring the method static

b. the type of the parameter position being declared as an int

✔ c. the value of the first parameter

d. the method having more than one parameter

e. the return type being declared as void

Correct Answer: c. the value of the first parameter

Explanation: Same as Q18 — if position is a very large or negative value beyond bounds, it crashes.

Q36. Which command can display the current ARP cache on a Windows system?

a. ipconfig

b. ping

✔ c. arp -a

d. netstat -r

Correct Answer: c. arp -a

Explanation: The 'arp -a' command displays the current ARP cache on Windows, showing IP-to-MAC address mappings.

Q37. In a graph, what does a cycle represent?

a. A linear path

b. A spanning tree

c. A set of disconnected vertices

✔ d. A path that starts and ends at the same vertex

Correct Answer: d. A path that starts and ends at the same vertex

Explanation: In graph theory, a cycle is a closed path — it starts and ends at the same vertex with no repeated edges.

Q38. Which of the following is a divide and conquer algorithm?

✔ a. Binary Search

b. Insertion Sort

c. Linear Search

d. Bubble Sort

Correct Answer: a. Binary Search

Explanation: Binary Search is a classic divide-and-conquer algorithm that repeatedly halves the search space.

Q39. Which of the following is true regarding the flow of execution in a while statement? (same as Q30)

a. If the condition is yielding not true, execute the body

✔ b. If the condition is false, exit the while statement and continue execution

c. All

d. Evaluates the condition yielding only True

Correct Answer: b. If the condition is false, exit the while statement and continue execution

Explanation: When the while condition evaluates to False, the loop exits and execution continues.

Q40. Which one of the following scheduling algorithms most likely to cause starvation? (same as Q32)

a. Round-Robin (RR)

b. First-Come, First Served (FCFS)

✔ c. Short-Job First (SJF)

d. None

e. Priority-Based With Aging

Correct Answer: c. Short-Job First (SJF)

Explanation: SJF is non-preemptive and always favors shorter jobs, which may starve longer ones indefinitely.

Q41. Which one of the following scheduling algorithms most likely to cause starvation? (same as Q32)

a. Round-Robin (RR)

b. First-Come, First Served (FCFS)

✔ c. Short-Job First (SJF)

d. None

e. Priority-Based With Aging

Correct Answer: c. Short-Job First (SJF)

Explanation: SJF without aging is the scheduling algorithm most prone to starvation of long processes.

Q42. Bereket needs to send an email to his supervisor with an attachment that includes sensitive information. (same as Q33)

a. Digital signature

✔ b. Encryption

c. Hashing

d. Data masking

Correct Answer: b. Encryption

Explanation: To maintain confidentiality of an email attachment, encryption ensures only the intended recipient can read it.

Q43. Consider the following setValue method — which one might cause an ArrayIndexOutOfBounds exception:

a. not declaring the method static

b. the type of the parameter position being declared as an int

✔ c. the value of the first parameter

d. the method having more than one parameter

e. the return type being declared as void

Correct Answer: c. the value of the first parameter

Explanation: If position is 0, list[-1] is accessed. If it's completely out of bounds, ArrayIndexOutOfBounds is thrown.

Q44. Which of the following best describes a compiled language?

a. Code is always faster in compiled languages because they have no errors

b. Code is executed line-by-line

✔ c. Code is translated directly into machine code before execution

d. Compiled languages cannot be used for low-level programming

Correct Answer: c. Code is translated directly into machine code before execution

Explanation: A compiled language translates source code entirely into machine code before execution.

Q45. Which principle refers to minimizing the interdependence between software modules?

a. Modularity

b. Cohesion

c. Abstraction

✔ d. Coupling

Correct Answer: d. Coupling

Explanation: Coupling refers to the degree of interdependence between modules. Low coupling is a desirable design goal.

Q46. _____ is a set of one or more attributes taken collectively to uniquely identify a record.

a. Primary Key

b. Foreign key

✔ c. Super key

d. Candidate key

Correct Answer: c. Super key

Explanation: A super key is any set of one or more attributes that uniquely identifies a record. A primary key is a minimal super key.

Q47. Consider the inheritance hierarchy shown in the diagram below: [Class inheriting diagram with two subclasses]. What does the arrow from subclasses to Class represent?

✔ a. Inheritance

b. Composition

c. Aggregation

d. Association

Correct Answer: a. Inheritance

Explanation: The hollow triangle/arrow pointing from a subclass to a superclass in UML class diagrams represents inheritance.

Q48. You are comparing different types of authentications. Of the following choices, which one uses multifactor authentication?

a. A system that requires users to enter a username and password

✔ b. A system that requires users to have a smart card and a PIN

c. A system that checks an employee's fingerprint and does a vein scan

d. A cipher door lock that requires employees to enter a code to open the door

Correct Answer: b. A system that requires users to have a smart card and a PIN

Explanation: Multifactor authentication (MFA) combines something you HAVE (smart card) with something you KNOW (PIN).

Q49. Which evaluation metric is most suitable for imbalanced classification problems?

a. Accuracy

b. R-squared

✔ c. F1-Score

d. Mean Squared Error (MSE)

Correct Answer: c. F1-Score

Explanation: F1-Score is the harmonic mean of precision and recall, making it ideal for imbalanced datasets.

Q50. In linear regression, the predicted value is a linear function of the _____.

a. distance from the hyperplane

✔ b. independent variables

c. class probabilities

d. dependent variable

Correct Answer: b. independent variables

Explanation: In linear regression, the predicted value is modeled as a linear function of the independent variables.

Q51. Your organization hired a cybersecurity expert to perform a security assessment. After running a vulnerability scan, she sees: Host IP 192.168.1.10 OS Apache httpd 2.433 Vulnerable to mod_auth exploit. However, she verified that the mod_auth module has not been installed or enabled on the server. Which of the following BEST explains this scenario?

✔ a. A false positive

b. The result of a credentialed scan

c. The result of a non-credentialed scan

d. A false negative

Correct Answer: a. A false positive

Explanation: A false positive occurs when a security tool reports a vulnerability that doesn't actually exist.

Q52. What is the main function of the three-way handshake in TCP?

a. Flow control

b. Authentication

c. Reliable data delivery

✔ d. Connection setup and synchronization

Correct Answer: d. Connection setup and synchronization

Explanation: The TCP three-way handshake (SYN, SYN-ACK, ACK) establishes a connection and synchronizes sequence numbers.

Q53. A class called Network has been declared with private attributes maxNumberOfUsers and currentNumberOfUsers. A class SubNet extends Network. In a program that declares an initializes an object sub of SubNet, which of the following statements would NOT cause a compiler error?

a. sub.maxNumberOfUsers = 100;

✔ b. sub.setMaxNumberOfUsers(100);

c. sub.currentNumberOfUsers = 50;

d. System.out.println(sub.maxNumberOfUsers);

Correct Answer: b. sub.setMaxNumberOfUsers(100);

Explanation: Private attributes are not accessible directly from outside the class. Only public mutator methods can be safely called.

Q54. Which one of the following mechanisms allows execution of multiple processes by switching CPU even if the currently executing process is not completed?

✔ a. Time-sharing

b. Multiprogramming

c. Multiprocessing

d. None

e. Monoprogramming

Correct Answer: a. Time-sharing

Explanation: Time-sharing allows the CPU to switch between processes rapidly, giving the illusion of parallel execution.

Q55. _____ involves generating, collecting, disseminating, and storing information.

a. Concurrent Management

b. Critical Management

c. Configuration Management

✔ d. Communication Management

Correct Answer: d. Communication Management

Explanation: Communication Management in project management involves generating, collecting, distributing, storing, and disposing of project information.

Q56. Chala is a security practitioner tasked with ensuring that the information on the organization's public website is not changed by anyone outside the organization. This task is an example of ensuring _____.

a. Availability

✔ b. Integrity

c. Confidentiality

d. Authentication

Correct Answer: b. Integrity

Explanation: Integrity ensures that information is not modified by unauthorized parties.

Q57. Which software process model emphasizes strict sequential flow between phases?

a. Agile Model

b. Incremental Model

c. Spiral Model

✔ d. Waterfall Model

Correct Answer: d. Waterfall Model

Explanation: The Waterfall model is a sequential model with distinct phases where each must complete before the next begins.

Q58. What is the output of the following code? >>> my_list = [1,2,3,2,4] >>> my_list.remove(2) >>> my_list

a. None

b. [1,2,2,4]

✔ c. [1,3,2,4]

d. [1,3,4]

Correct Answer: c. [1,3,2,4]

Explanation: list.remove(2) removes only the FIRST occurrence of 2.

Q59. Which of the following best describes the role of a file system in an operating system?

a. It manages the computer's hardware resources, such as CPU and memory

✔ b. It provides a structured way to store, organize, retrieve, and manage files on storage devices

c. It optimizes network bandwidth usage for file transfers over the internet

d. It handles user authentication and access control for logging into the system

e. None

Correct Answer: b. It provides a structured way to store, organize, retrieve, and manage files on storage devices

Explanation: A file system manages how data is stored on physical storage storage devices.

Q60. Delphi approach is also called _____ group consensus technique.

a. Fully Consultative

b. Non-Consultative

✔ c. Consultative

d. Partially Consultative

Correct Answer: c. Consultative

Explanation: The Delphi technique is a consultative group consensus method where experts give anonymous opinions iteratively.

Q61. What is the output of the following code? >>> my_list = [1,2,3,4,5,6,7,8,9,10] >>> my_list[5::-2]

✔ a. [6,4,2]

b. [5,7,9]

c. [5,4,3,2,1]

d. []

Correct Answer: a. [6,4,2]

Explanation: my_list[5::-2] starts at index 5 (value 6), steps backward by 2: values 6, 4, then 2.

Q62. _____ is a hierarchical and incremental decomposition of the project into phases, deliverables and work packages.

a. Work Integration Structure

b. Task Integration Structure

c. Bottom-up Integration Structure

✔ d. Work Breakdown Structure

Correct Answer: d. Work Breakdown Structure

Explanation: The Work Breakdown Structure (WBS) is a hierarchical decomposition of a project's total scope.

Q63. The bias-variance trade-off implies that:

a. Increasing model complexity reduces both bias and variance

b. Underfitting is caused by low variance

✔ c. Simple models have high bias and low variance

d. Overfitting occurs when bias is too high

Correct Answer: c. Simple models have high bias and low variance

Explanation: The bias-variance trade-off shows that simple/underfit models have high bias and low variance.

Q64. What is the primary objective of virtual memory in a computer system?

a. increase the physical memory capacity

b. None

c. increase execution speed by catching frequently used data

✔ d. allow processes use more memory than physical available

e. secure part of memory from being accessed by other processes

Correct Answer: d. allow processes use more memory than physical available

Explanation: Virtual memory extends effective memory by using disk space as swap space.

Q65. What is printed by the following statements? >>> str = 'This is python code' >>> print(str[2] + str[-6])

a. tn

✔ b. io

c. in

d. to

Correct Answer: b. io

Explanation: str[2] = 'i' and str[-6] = 'o'. Combining them yields 'io'.

Q66. Sara is a database manager. Sara is allowed to add new users to the database, remove current users, and create new usage functions for the users. Sara is not allowed to read the data in the fields of the database itself. This is an example of:

✔ a. Role Based Access Control

b. Discretionary Access Control

c. Mandatory Access Control

d. Behavior Based Access Control

Correct Answer: a. Role Based Access Control

Explanation: RBAC assigns permissions based on organizational roles rather than personal identities.

Q67. Which type of data can be stored in the database?

a. Text, files containing data

✔ b. All of the above

c. Image oriented data

d. Data in the form of audio or video

Correct Answer: b. All of the above

Explanation: Modern relational databases can store text, images, structured files, audio, and video binary objects.

Q68. What is the primary function of an Intent in Android app navigation?

a. To manage background threads for long-running operations

✔ b. To request an action from another app component

c. To store application settings persistently

d. To define the visual appearance of a screen

Correct Answer: b. To request an action from another app component

Explanation: An Intent is a core messaging object used to request an action from another app component.

Q69. What is the primary purpose of WAI-ARIA?

a. To improve website SEO rankings

✔ b. To make web applications more accessible

c. To increase website loading speed

d. To replace HTML with a new accessibility standard

Correct Answer: b. To make web applications more accessible

Explanation: WAI-ARIA provides specific accessibility semantic attributes to improve rich client dynamic web components.

Q70. A program is required to add two integer numbers together. The GUI for this program shows two text fields and a Calculate button. What is the component labeled with an arrow in the diagram?

a. JFrame

✔ b. JTextField

c. JLabel

d. JButton

Correct Answer: b. JTextField

Explanation: The text entry inputs used inside basic Java Swing applications correspond directly to the JTextField component class.

Q71. What is meant by software quality in software engineering? (same as Q5)

a. Complexity of code

b. High number of features

✔ c. Degree to which software meets requirements and is free of defects

d. Low cost

Correct Answer: c. Degree to which software meets requirements and is free of defects

Explanation: Software quality is defined as compliance to operational rules and absence of coding bugs.

Q72. The specific set of sequential tasks upon which the project completion date depends or the longest full path is called _____.

a. Full path

b. Project Path

✔ c. Critical Path

d. Complete Path

Correct Answer: c. Critical Path

Explanation: The Critical Path is the longest continuous chain of dependent tasks running through a project network diagram.

Q73. Which of the following sorting algorithms has the best average-case time complexity?

✔ a. Quick Sort

b. Bubble Sort

c. Insertion Sort

d. Selection Sort

Correct Answer: a. Quick Sort

Explanation: Quick Sort maintains an exceptionally performant average-case execution profile of O(n log n).

Q74. What is the worst-case time complexity of Merge Sort?

a. $O(n^2)$

b. O(log n)

✔ c. O(n log n)

d. O(n)

Correct Answer: c. O(n log n)

Explanation: Merge Sort maintains an explicit structural execution bound of O(n log n) across best, average, and worst cases.

Q75. How do you check if an element exists in a set? Using

✔ a. in keyword

b. exists()

c. contains()

d. check()

Correct Answer: a. in keyword

Explanation: Python uses the highly readable, built-in 'in' operator to perform set evaluation checks cleanly.

Q76. Which of the following is an example of a 'Something you know' authentication factor?

a. GPS

b. Finger Printing

✔ c. Password

d. ATM Card

Correct Answer: c. Password

Explanation: Passwords represent a knowledge factor configuration matching the criteria of 'something you know'.

Q77. What is the output of this expression: >>> (2 + 5) ** 2 // 10 + 2 * 3

a. 12.0

✔ b. 10

c. 8

d. 12

Correct Answer: b. 10

Explanation: Evaluating order: (2+5)**2 = 49. Floor division: 49 // 10 = 4. Multiplication: 2 * 3 = 6. Add results: 4 + 6 = 10.

Q78. A technique to prevent modification anomalies by creating a different class of relations in RDMS are called?

a. Functional Dependencies

✔ b. Normal Forms

c. Referential integrity

d. Data integrity

Correct Answer: b. Normal Forms

Explanation: Normalization techniques eliminate redundancy anomalies across table relationships explicitly.

Q79. Which operation is used to extract specified columns from a table?

a. Substitute

b. Extract

✔ c. Project

d. Join

Correct Answer: c. Project

Explanation: Relational algebra uses the project projection operator (π) to select specific column targets from tables.

Q80. A team is designing a weather app that must work offline in remote areas but also sync data when internet connectivity is available. Which architectural approach addresses this need?

a. Only online-based architecture

b. Peer-to-peer architecture

✔ c. Client-side caching with periodic synchronization

d. Pure serverless architecture (no offline support)

Correct Answer: c. Client-side caching with periodic synchronization

Explanation: This approach keeps data locally accessible when networks drop and updates the cloud automatically upon connection.

Q81. Which of the following is the primary goal of software engineering?

✔ a. To produce reliable, efficient, and maintainable software

b. To develop low-level machine code

c. To increase hardware performance

d. To create visually appealing user interfaces

Correct Answer: a. To produce reliable, efficient, and maintainable software

Explanation: The core foundational mission of Software Engineering focuses on robust execution alongside predictable maintenance frameworks.

Q82. Which of the following is a divide and conquer algorithm? (same as Q38)

✔ a. Binary Search

b. Insertion Sort

c. Linear Search

d. Bubble Sort

Correct Answer: a. Binary Search

Explanation: Binary search uses a classical binary search space reduction tree strategy matching divide-and-conquer architectures.

Q83. In the context of search problems, what does a heuristic function do?

✔ a. Estimates the cost from a node to the goal

b. Calculates the actual cost of a path

c. Calculates the cost from start to goal

d. Finds the goal state directly

Correct Answer: a. Estimates the cost from a node to the goal

Explanation: Heuristic computations generate dynamic evaluations estimating remaining cost margins to goal states.

Q84. How do you create a CSS rule that applies only to screen devices with a maximum width of 600px?

a. @media screen and (min-width: 600px) { ... }

b. @media print screen and (max-width: 600px) { ... }

✔ c. @media screen and (max-width: 600px) { ... }

d. @media all and (max-width: 600px) { ... }

Correct Answer: c. @media screen and (max-width: 600px) { ... }

Explanation: The screen flag targets display panels while max-width applies responsive style limits down to exactly 600px targets.

Q85. What is the main advantage of using kernel methods in machine learning?

a. They are faster than other machine learning algorithms for all tasks

b. They can only be used with Support Vector Machines (SVMs)

c. They require less data for training compared to other methods

✔ d. They allow linear algorithms to work effectively with non-linear data

Correct Answer: d. They allow linear algorithms to work effectively with non-linear data

Explanation: Kernel transforms project observations implicitly into higher dimensions where separating hyperplanes become linear.

Q86. Which CSS property is used to control the stacking order of positioned elements?

a. display

b. position

c. float

✔ d. z-index

Correct Answer: d. z-index

Explanation: CSS utilizes the explicit coordinate depth property z-index to coordinate vertical stacking logic rules.

Q87. How can web developers accommodate users with mobility impairments?

✔ a. Ensuring all controls are accessible via keyboard

b. Making users rely on a mouse for navigation

c. Limiting access to only touch-screen devices

d. Removing all interactive elements

Correct Answer: a. Ensuring all controls are accessible via keyboard

Explanation: Keyboard compliance layouts remove dependencies on pointer interactions to help access goals.

Q88. Consider class ClassA (with testMethod printing 'Hello') and class ClassB extends ClassA (with testMethod printing 'Goodbye'). If a ClassB object calls testMethod(), what is printed?

a. Hello

✔ b. Goodbye

c. Hello then Goodbye

d. Goodbye then Hello

Correct Answer: b. Goodbye

Explanation: Polymorphism ensures dynamic method dispatch resolves directly to the object instance's concrete execution method.

Q89. Your organization hired a cybersecurity expert. After a vulnerability scan: Host IP 192.168.1.10 OS Apache httpd 2.433 Vulnerable to mod_auth exploit. She verified that mod_auth has NOT been installed. Which BEST explains this scenario?

a. The result of a credentialed scan

b. A false negative

c. The result of a non-credentialed scan

✔ d. A false positive

Correct Answer: d. A false positive

Explanation: When software alerting configurations flags vulnerability alerts against absent missing system extensions, a false positive occurs.

Q90. What is printed by the following statements? >>> Str1 = 'PYTHON programming' >>> print(Str1[2] * Str1.index('O'))

a. OOOO

b. THON

✔ c. TTTT

d. YYYY

Correct Answer: c. TTTT

Explanation: Index 2 locates string value 'T'. The target parameter method resolves character 'O' index down to integer 4. 'T' repeated 4 times yields 'TTTT'.

Q91. What is the output of the following code? >>> my_list = [1,2,3,4,5,6,7,8,9,10] >>> my_list[5::-2] (same as Q61)

✔ a. [6,4,2]

b. [5,7,9]

c. [5,4,3,2,1]

d. []

Correct Answer: a. [6,4,2]

Explanation: Slicing runs backward from target element index 5 stepping by intervals of -2 across array spaces.

Q92. Which statement about ensemble methods (e.g., Random Forests) is TRUE?

a. They require all base models to be identical

b. They perform poorly compared to single models

c. They combine weak learners to reduce bias only

✔ d. They average predictions to decrease variance

Correct Answer: d. They average predictions to decrease variance

Explanation: Ensemble random generation constructs group voting structures to normalize error boundaries across training outputs.

Q93. What is the primary purpose of the RecyclerView widget?

a. To display a single image on the screen

✔ b. To display lists of data by efficiently reusing item views

c. To provide a simple button with click functionality

d. To manage background tasks efficiently

Correct Answer: b. To display lists of data by efficiently reusing item views

Explanation: RecyclerView structural loops optimize layout processing overhead by cycling view object rows directly.

Q94. When you want to navigate to a specific Activity within your application, which type of Intent is typically used?

✔ a. Explicit intent

b. Implicit intent

c. Pending intent

d. Broadcast intent

Correct Answer: a. Explicit intent

Explanation: Explicit intent requests name context target targets inside internal application architectures directly.

Q95. What is the purpose of the box-sizing property in CSS?

✔ a. To include padding and border in the element's total size

b. To set the total margin of the box

c. To define the size of the total content and margin box

d. To apply a shadow and margin to the box

Correct Answer: a. To include padding and border in the element's total size

Explanation: Applying border-box layout rules wraps sizing values over calculation parameters directly.

Q96. Your organization's backup policy for a file server dictates that the amount of time needed to restore backups should be minimized. Which of the following backup plans would BEST meet this need?

a. Full backups on Sunday and incremental backups on the other six days of the week

✔ b. Full backups on Sunday and differential backups on the other six days of the week

c. Differential backups on Sunday and incremental backups on the other six days of the week

d. Incremental backups on Sunday and differential backups on the other six days of the week

Correct Answer: b. Full backups on Sunday and differential backups on the other six days

Explanation: Differential strategies mean retrieval sequences require only the initialization baseline image coupled with the most recent delta package.

Q97. In a neural network, the activation function introduces:

a. Regularization to prevent overfitting

✔ b. Non-linearity to model complex patterns

c. Dimensionality reduction

d. Linear decision boundaries

Correct Answer: b. Non-linearity to model complex patterns

Explanation: Activation mechanisms break strict linear matrices constraints, enabling deep architectures to approach intricate boundaries.

Q98. A _____ is the lowest level of work on the project.

a. Task Set

✔ b. Task

c. Milestone

d. Work product

Correct Answer: b. Task

Explanation: Work Breakdown frameworks isolate execution nodes down to elementary assignable chunks called tasks or work packages.

Q99. Return of Investment (ROI) ROI = _____

a. (Gain - Cost)/Gain

b. (Cost - Gain)/Gain

✔ c. (Gain - Cost)/Cost

d. (Cost - Gain)/ Cost

Correct Answer: c. (Gain - Cost)/Cost

Explanation: Financial evaluation structures balance absolute net revenue margins over total initial outlays directly.

Q100. A startup is building an e-commerce platform expecting rapid growth. They need an architecture that allows independent scaling of the product catalog, payment processing, and recommendation engine. Which architectural style is most suitable?

a. Single-tier architecture (all logic on the client side)

b. Monolithic architecture (all components in one codebase)

c. Shared-database architecture (all services use one database)

✔ d. Microservices architecture (independent, loosely coupled services)

Correct Answer: d. Microservices architecture (independent, loosely coupled services)

Explanation: Decoupled component arrays permit individual vertical service nodes to expand or change operations contextually without crashing surrounding modules.

Q101. Slack = _____

a. LS-EF

b. ES-LF

c. ES-LS

✔ d. LF-EF

Correct Answer: d. LF-EF

Explanation: Total activity float values represent differences mapped cleanly across finish variations like Latest Finish minus Earliest Finish.

Q102. Which of the following is the correct syntax to create a function in JavaScript?

a. function: myFunction() {}

b. function = myFunction() {}

c. function => myFunction() {}

✔ d. function myFunction() {}

Correct Answer: d. function myFunction() {}

Explanation: Standard JavaScript structural declarations require explicit function tagging declarations matching parameter parentheses.

Q103. Which of the following function declarations is correct to define a function that accepts a variable number of arguments?

✔ a. def func(*args):

b. Not possible in python

c. def func(args*):

d. def func(**args):

Correct Answer: a. def func(*args):

Explanation: Single pointer star notation unpacks optional extra matching sequence inputs inside functional inputs as standard tuples.

Q104. Which of the following is a key challenge in reinforcement learning?

a. Requirement of perfectly balanced datasets

✔ b. The trade-off between exploration and exploitation

c. Lack of labeled training data

d. High computational cost of matrix operations

Correct Answer: b. The trade-off between exploration and exploitation

Explanation: Agents balance finding alternative potential reward avenues over executing current optimized state actions.

Q105. Which algorithm is most appropriate when the cost of actions varies and finding the least-cost solution is important?

a. Depth-First Search

b. Breadth-First Search

c. Greedy Best-First Search

✔ d. Uniform Cost Search

Correct Answer: d. Uniform Cost Search

Explanation: Uniform cost graphs build out optimal structural routes across varying edge values step-by-step.

Q106. How can you apply a CSS Grid layout to a container?

a. display: block;

✔ b. display: grid;

c. display: inline-grid;

d. display: flex;

Correct Answer: b. display: grid;

Explanation: Declaring structural grid configurations turns structural elements instantly into functional grid parent layers.
